

Box 30.1 Alterations of Plasma Protein and Procoagulant Factors in Nephrotic Syndrome

- Plasma globulins**

 - Decreased immunoglobulin G (IgG)
 - Normal or increased IgA, IgM, or IgE
 - Increased α2- and β-globulins
 - Decreased α1-globulins

Metal- and hormone-binding proteins

 - Loss of metal binding proteins: iron, copper, zinc
 - Loss of erythropoietin, transferrin
 - Loss of thyroid-binding globulin, vitamin D-binding protein, sex-hormone binding globulin, transcortin

Complement system

 - Decreased factor B and I
 - Decreased C1q, C2, C8, C9
 - Increased C3
 - Normal C1s, C4, and C1 inhibitor
- Coagulation components**

 - Increased factors II, V, VII, VIII, X, XIII
 - Decreased factors IX, XI, XII
 - Increased plasma fibrinogen
 - Increased intravascular fibrin formation
 - Decreased fibrinolytic activity
 - Decreased plasma plasminogen
 - Increased tissue type plasminogen activator
 - Increased α2-macroglobulin
 - Decreased α1-antitrypsin
 - Decreased antithrombin III
 - Decreased protein S

Other procoagulant factors

 - Increased blood viscosity
 - Hemoconcentration
 - Thrombocytosis
 - Increased platelet aggregability

Box_30_1

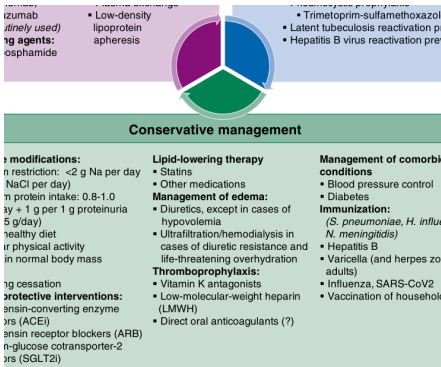


Fig. 30.2 Overview of management of patients with nephrotic syndrome.

Fig_30_2



Fig. 30.5 (A) Unremarkable light microscopy appearance of a biopsy specimen from a patient with minimal change disease. Glomerular basement membranes are thick, and there is no glomerular hypercellularity or mesangial matrix expansion. Urinary mesangial silver stain, ×500. (B) Electron micrograph of a glomerular capillary wall from a patient with minimal change disease showing extensive podocyte foot process effacement (arrows) and microvilli projections of podocyte foot processes (arrowheads). ×5000.

Fig_30_5

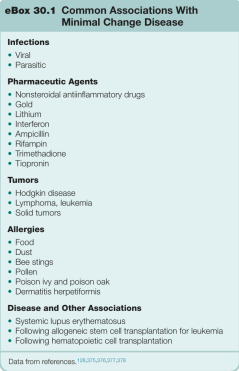


Fig. 30.3 Management of edema and hypovolemia in nephrotic syndrome. In children, edema severity is graded empirically as follows: mild (<7% increase in body weight), moderate (7% to 15%), and severe (>15%). No grading system has been established for adult patients. (Adapted from Sinha A, Bagga A, Banerjee S, et al. Steroid sensitive nephrotic syndrome: revised guidelines. Indian Pediatr. 2021;58:461-461.)

Fig_30_3

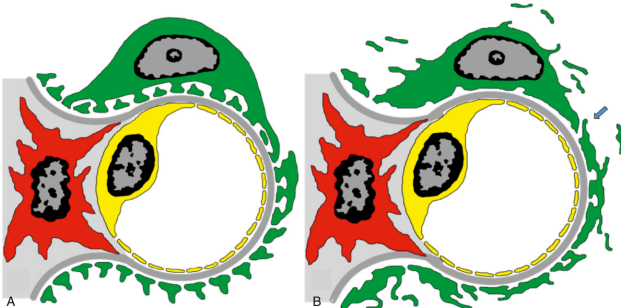


Fig. 30.6 Diagrammatic representations of the ultrastructural features of a normal glomerular capillary loop (A) and a capillary loop with features of minimal change disease (B). The latter has effacement of podocyte foot processes and microvilli projections of podocyte cytoplasm. (Courtesy J. C. Jennette.)

Fig_30_6

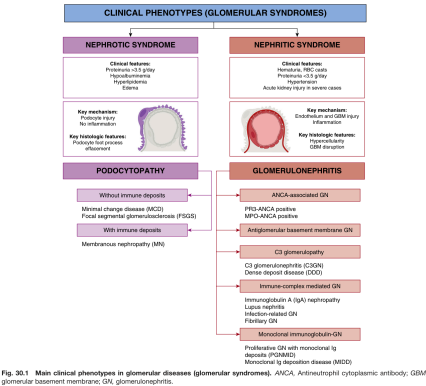


Fig. 30.1 Main clinical phenotypes in glomerular diseases (glomerular syndromes). ANCA, Antineutrophil cytoplasmic antibody; GBM, glomerular basement membrane; GN, glomerulonephritis.

Fig_30_1

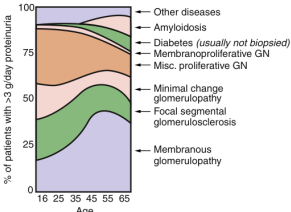


Fig. 30.4 Graphic depiction of frequencies of different forms of glomerular disease identified in kidney biopsy specimens from patients with proteinuria of more than 3 g of protein/day evaluated at the University of North Carolina Nephropathology Laboratory. Some diseases that cause proteinuria are underrepresented because they are not always evaluated by kidney biopsy. For example, in many patients, steroid-responsive proteinuria is given a presumptive diagnosis of minimal change disease and patients do not undergo biopsy, and most patients with diabetes and proteinuria are presumed to have diabetic glomerulosclerosis and do not undergo biopsy. GN, Glomerulonephritis.

Fig_30_4

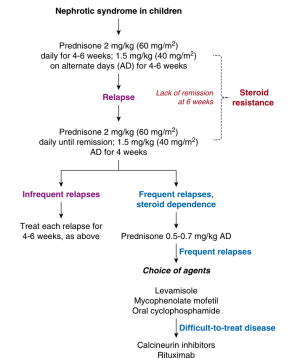


Fig. 30.7 Immunosuppressive treatment in nephrotic syndrome in children.

Fig_30_7

30. Nephrotic Syndrome_Primary and Secondary · assets 10-18 / 29

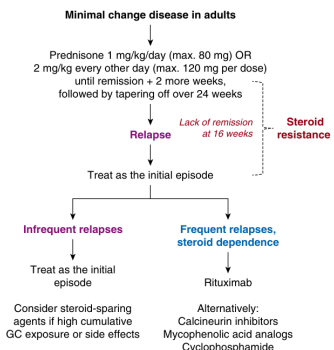


Fig. 30.8 Immunosuppressive treatment in minimal change disease in adults.

Fig_30_8

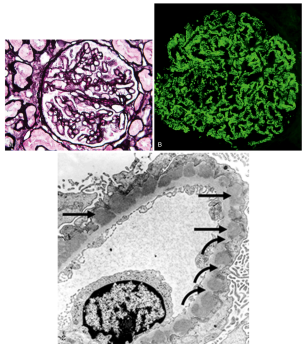
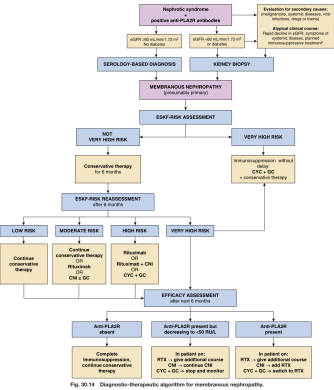
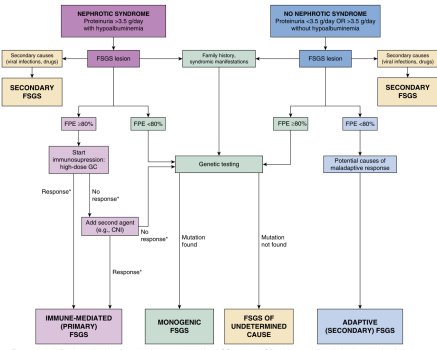


Fig. 30.11 (A) Light micrograph of a glomerulus with features of membranous nephropathy demonstrating apical, capillary, and other irregularities. (B) Electron micrograph showing features of membranous nephropathy including subendothelial immune deposits, mesangial electron-dense deposits, and podocyte foot process effacement.

Fig_30_11

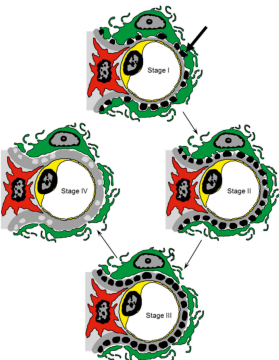


Fig_30_14



* Response is defined as reduction of proteinuria >50% in 8 weeks of GC treatment OR at least partial remission in 16 weeks of GC treatment.

Fig_30_9



Fig_30_12

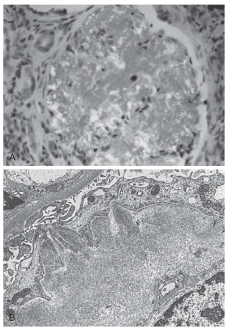


Fig. 30.15 Amyloidosis. (A) Congo red stain of a glomerulus that is largely replaced by amyloid demonstrating the characteristic birefringence under polarized light (x450). (B) Amyloidosis. Electron micrograph showing extensive infiltration of the glomerular basement membrane by 10-nm fibrils that project toward the urinary space (x8000).

Fig_30_15

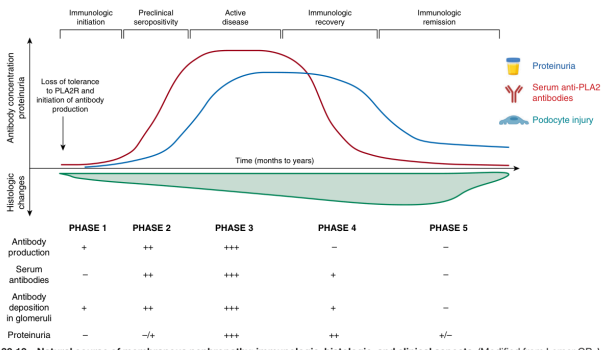


Fig. 30.10 Natural course of membranous nephropathy: immunologic, histologic, and clinical aspects. (Modified from Lerner GB, Virmani S, Henderson JM, Francis JM, Beck LH. A conceptual framework linking immunology, pathology, and clinical features in primary membranous nephropathy. *Kidney Int.* 2021;100:289-300.)

Fig_30_10

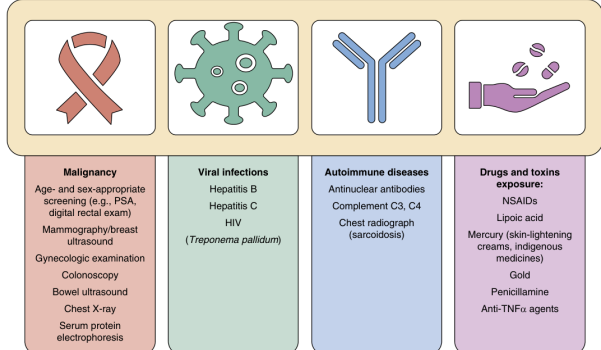


Fig. 30.13 Diagnostic evaluation for secondary causes of membranous nephropathy.

Fig_30_13

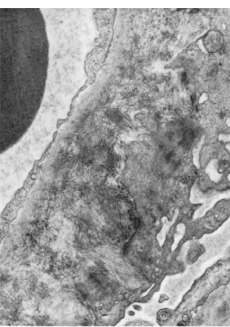
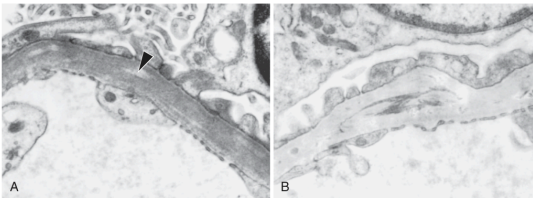


Fig. 30.16 Fibrillary glomerulonephritis. Electron micrograph showing the characteristic randomly oriented fibrils, measuring 16 to 20 nm within the glomerular basement membrane. The foot processes are effaced (x8000).

Fig_30_16

Fig_30_17

Table_30_2

Fig_30_18

Table_30_3Table_30_1Table_30_4Table_30_5Table_30_6Table_30_7

